Open Datasets for SysML and SysML v2:

A Comprehensive Survey of Research Resources

Survey Report

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Abstract

Model-Based Systems Engineering (MBSE) relies on the Systems Modeling Language (SysML) to capture system requirements, architecture, behavior, and parametrics in a consistent formalism. The newer **SysML v2** introduces a textual notation and richer semantics. As researchers explore ways to build and analyze systems using large language models, publicly available training data remains limited. This survey provides a comprehensive overview of open, research-usable datasets and model libraries for both **SysML v1.x** and **SysML v2**, documenting their key features, formats, application domains, and associated research contributions.

1 Introduction

The evolution of Model-Based Systems Engineering (MBSE) has been significantly influenced by the development and adoption of the Systems Modeling Language (SysML). While SysML v1.x established a solid foundation for systems modeling, SysML v2 represents a paradigm shift with its textual notation and enhanced semantic capabilities.

The intersection of MBSE with artificial intelligence, particularly large language models (LLMs), presents unprecedented opportunities for automated system design and analysis. However, progress in this domain is constrained by the scarcity of high-quality, publicly available training datasets. This survey addresses this gap by cataloging existing open datasets and repositories that can support research in automated systems modeling, model analysis, and AI-driven MBSE applications.

2 Open Datasets and Repositories

The following comprehensive table summarizes the current landscape of open datasets and repositories available for SysML and SysML v2 research. Each entry is characterized by its organizational source, temporal scope, key technical features, supported formats, application domains, and associated scholarly contributions.

Dataset/Repository	**Organization/Authors**	**Year & Version**	**Key Features**	**Formats**	**Domain/Examp**	**Asso Pub**
lightgray **SysML DELS Model Libraries**	NIST (Barbau & Bock)	2019 (SysML 1.4)	Discrete-event logistics system libraries; modular block architecture; requires MagicDraw 18.5+	XMI, HTML	Manufacturing plants, warehouses, supply chain optimization	NIST

Dataset/Repository	Organization/Authors	Year & Version	Key Features	Formats	Domain/Examples	Asso Pub
Wireless Factory Work-cell Model	NIST (Bock & Barbau)	2024	Structural component models, interface definitions, parametric constraints for wireless industrial systems	MagicDraw XML, HTML	Industrial wireless connectivity, IoT manufacturing	NIST docu
lightgray **SysPhS Physical-Interaction Models**	NIST (Manion & Bock)	2024 (SysPhS 1.0)	Libraries for rotational/translational mechanics, heat transfer; 1-D physics simulation; Modelica/Simscape translation	SysML/SysPhS, executable code	Manufacturing physical interactions, mechatronics	NIST
SysLMA Translator	NIST (Barbau & Bock)	2025	SysML model translation with SysLMA profile extension; logistics analysis integration; comprehensive examples	Source & binary code, model libraries	Logistics analysis, supply chain modeling	NIST
lightgray **SysPhS Translator v1.0**	NIST (Manion & Bock)	2019–2020	SysPhS 1.0 implementation; SysML+SysPhS to Modelica/Simulink translation; cross-platform compatibility	Source & binary code, example models	Cross-platform simulation, multi-physics modeling	Data docu
SysPhS Translator v1.1	NIST (Manion & Bock)	2023–2025	Enhanced SysPhS 1.1 implementation; improved Modelica & Simulink/Simscape generation; expanded model library	Source & binary code, sample models	Advanced simulation platforms, model interoperability	NIST docu page
lightgray **OBM & SysML→Alloy Translations**	NIST (Manion & Bock)	2024–2025	Alloy Analyzer integration; SysML behavioral model translation; Ontological Behavior Modeling; executability verification	Alloy code, verification examples	Behavior verification, model consistency checking	NIST 8388

Dataset/Repository	Organization/Authors	Year & Version	Key Features	Formats	Domain/Examples	Asso Pub
SysML v2 Models Collection	GfSE Community	2025 (ongoing)	Curated high-quality SysML v2 models; best-practice examples; educational resources; LLM training data; BSD-3 license	.sysml, Markdown documentation	Cross-domain modeling, educational applications	Repo docu
lightgray **SysML v2 Release Repository**	OMG Systems Modeling Community	2023–2025	Official SysML v2 specifications; normative libraries; Eclipse & Jupyter integration; comprehensive tooling support	PDF, .sysml/.kerml, XMI	General systems modeling, standards compliance	Offic speci docu
SysMBench Benchmark	Jin et al.	2025	151 curated scenarios; natural-language requirements; textual & graphical models; domain classification; difficulty grading	SysML text & diagrams, metadata	Multi-domain (automotive, aerospace, photography, traffic)	SysM resea
lightgray **Aerospace SysML Model Dataset**	Zhang & Yang	2024	Domain-specific SysML models; validation rule sets; spacecraft system knowledge; MBSE recommendation training data	SysML models, validation rules	Spacecraft systems, aerospace engineering	Appl jour

Table 1: Comprehensive overview of open SysML and SysML v2 datasets and repositories

3 Research Impact and Applications

The datasets cataloged above have enabled significant research contributions across multiple domains of systems engineering and artificial intelligence. The following subsections detail the key research outcomes and their implications for the broader MBSE community.

3.1 NIST Research Contributions

The National Institute of Standards and Technology (NIST) has been instrumental in developing foundational resources for MBSE research:

- **NIST IR 8262** documents the Discrete Event Logistics Systems (DELS) libraries, providing reference implementations for logistics and manufacturing systems analysis with comprehensive validation studies.

- **NIST IR 8490** introduces expanded physical-interaction component libraries for SysPhS, demonstrating cross-platform simulation capabilities and providing empirical validation of translation accuracy.
- **NIST IR 8571** defines the SysLMA (Systems Logistics Modeling and Analysis) extension methodology, showcasing practical applications in supply chain optimization and logistics workflow automation.
- **NIST IR 8388-upd1** proposes novel verification methodologies for SysML behavioral models through formal translation to Alloy constraints, addressing critical gaps in model executability verification.

3.2 Community-Driven Initiatives

3.2.1 SysML v2 Ecosystem Development

The **SysML v2 Release repository** serves as the authoritative source for the new modeling standard, providing:

- Official specification documents with detailed semantic definitions
- Normative libraries serving as reference implementations
- Comprehensive tooling infrastructure including Eclipse and Jupyter integrations
- Extensive example models demonstrating best practices

The **GfSE community collection** complements official resources by providing curated, high-quality models specifically designed for educational applications and LLM training, addressing the critical need for diverse, well-documented training data.

3.2.2 AI and Machine Learning Applications

Recent research has begun exploring the intersection of MBSE and artificial intelligence:

- **SysMBench** (Jin et al., 2025) represents the first comprehensive benchmark for evaluating large language models on SysML generation tasks. The study evaluated 17 different LLMs and introduced the SysMEval metric for assessing model quality, establishing baseline performance measurements for future research.
- **Zhang & Yang (2024)** developed domain-specific datasets for aerospace applications, demonstrating how specialized model collections can improve MBSE tool recommendations and modeling efficiency in specific engineering domains.

4 Dataset Characteristics and Quality Assessment

4.1 Format Diversity and Interoperability

The surveyed datasets exhibit significant diversity in format support, reflecting the evolving landscape of MBSE tooling:

- **Legacy formats**: XMI and HTML remain prevalent for SysML v1.x compatibility
- **Native textual formats**: .sysml and .kerml files represent the future of human-readable modeling

- **Executable representations**: Source code and binary distributions enable practical application
- **Documentation formats**: Markdown and PDF provide essential context and usage guidelines

4.2 Domain Coverage Analysis

The current dataset landscape demonstrates broad domain coverage:

- **Manufacturing and Logistics**: Well-represented through NIST contributions
- **Aerospace and Defense**: Emerging coverage through specialized datasets
- **Cross-domain Applications**: SysMBench provides multi-domain scenarios
- **Physical Systems**: Strong representation through SysPhS libraries

4.3 Licensing and Accessibility

Most datasets adopt open licensing models (BSD-3, LGPL/GPL) that facilitate research applications, though researchers should verify current licensing terms before use in derivative works.

5 Challenges and Future Directions

5.1 Current Limitations

Despite significant progress, several challenges persist in the current dataset ecosystem:

- **Scale limitations**: Most datasets remain relatively small-scale compared to datasets in other AI domains
- **Quality variability**: Inconsistent documentation and validation across different sources
- **Domain imbalance**: Some engineering domains remain underrepresented
- **Temporal coverage**: Limited longitudinal data for studying model evolution

5.2 Emerging Opportunities

The MBSE research community is positioned to address these limitations through:

- **Collaborative curation**: Community-driven efforts like the GfSE collection demonstrate the potential for collaborative dataset development
- **AI-assisted generation**: Large language models could help generate synthetic training data to augment existing collections
- **Industry partnerships**: Collaborations with industrial partners could provide larger-scale, real-world datasets
- **Cross-domain integration**: Efforts to create unified datasets spanning multiple engineering domains

6 Recommendations for Researchers

Based on this survey, we provide the following recommendations for researchers working in MBSE and related fields:

6.1 For Model Analysis Research

- Utilize NIST SysPhS libraries for physical system modeling validation
- Leverage OBM/Alloy translations for formal verification studies
- Consider SysML v2 Release repository for standards-compliant research

6.2 For AI and LLM Applications

- Start with SysMBench for baseline LLM evaluation
- Utilize GfSE community models for training data diversity
- Consider domain-specific datasets (aerospace) for specialized applications

6.3 For Tool Development

- Use SysML v2 Release repository for reference implementations
- Leverage NIST translators for interoperability validation
- Consider SysLMA extensions for logistics applications

7 Conclusion

The MBSE research community has made substantial progress in assembling open datasets to support research on model analysis, simulation, and large language model training. NIST's comprehensive contributions provide robust foundations for SysML v1.x research, while emerging SysML v2 resources establish the groundwork for next-generation modeling research.

The introduction of benchmarks like SysMBench and domain-specific datasets represents crucial steps toward systematic evaluation of AI applications in systems engineering. However, significant opportunities remain for expanding dataset scale, improving quality consistency, and broadening domain coverage.

As the field continues to evolve, the success of AI-driven MBSE applications will largely depend on the availability of high-quality, diverse, and well-documented datasets. The resources cataloged in this survey provide an essential foundation, but sustained community effort will be required to build the comprehensive data ecosystem needed to realize the full potential of intelligent systems engineering tools.

The convergence of traditional MBSE methodologies with modern AI capabilities promises to transform how complex systems are designed, analyzed, and maintained. The datasets surveyed here represent the first steps in this transformation, providing the empirical foundation necessary for evidence-based advances in automated systems engineering.